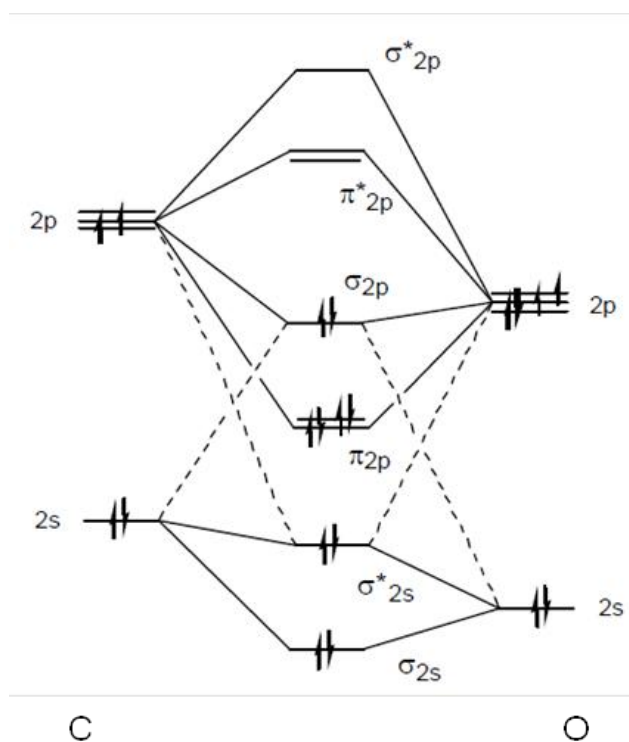


- 1) Write the noble gas ground-state electron configurations for the following atoms and ions (**5 points**):
(a) Mg; (b) N; (c) P; (d) Eu^{3+} ; (e) Ni^{2+} .
- 2) Using Slater's rules calculate the effective nuclear charge of the electrons in the following orbitals (**10 points**): (a) $3s$ in a magnesium atom; (b) $2p$ in a nitrogen atom; (c) $3p$ in a phosphorus atom; (d) $4f$ in a europium Eu^{3+} ion, (e) $3d$ in a nickel Ni^{2+} ion.
- 3) Answer to the following questions and explain your reasoning, also by using your answers to the previous exercise (**4 points**):
- Which is the largest atom between magnesium and phosphorus?
 - Which is the most electronegative between nitrogen and phosphorus?
- 4) By taking in consideration the position of the elements in the periodic table, indicate the true sentences by a cross (**9 points**):
- ☐ Europium (Eu) has higher mass than hafnium (Hf)
 - ☐ Magnesium has smaller atomic radius than sodium
 - ☐ The first ionization energy is higher for Mg than for Na
 - ☐ The first ionization energy is higher for Na than for Cs
 - ☐ Mg has higher second ionization energy than Na
 - ☐ The attachment of one electron to a neutral Li atom is more favorable than to a neutral Be atom.
 - ☐ In absolute values F has higher electron affinity than Li.
 - ☐ Fluorine is more electronegative than iodine.
 - ☐ $3+$ is the most common oxidation state for aluminum
- 5) Draw the Lewis structure of SnCl_3^- . What is the bond order of this molecule? Determine the shape of this molecule by using the VSEPR model? What is the hybridization according to the Valence Bond model for this molecule (**4 points**)?
- 6) Draw the Lewis structure of NH_2^- . What is the bond order of this molecule? Determine the shape of this molecule by using the VSEPR model? What is the hybridization according to the Valence Bond model for this molecule (**4 points**)?

- 7) Supposing that nitric oxide (NO) has a molecular orbital structure similar to carbon monoxide (CO) see below, write the molecular orbital diagram and define the bond order for NO (7 points).



- 8) Use the molecular orbital theory to predict the bond order of the O_2^{2-} ion (7 points).